

Practice Quiz: Planning — Your Brain Is a Time Machine

Part A: Multiple Choice

1. Planning in active inference is your brain's ability to: A) Perfectly predict the future B) Simulate possible futures and choose the path most likely to get you what you want C) Avoid all risk D) Do nothing until you're forced to act

Answer: B — Your brain runs simulations of different possible futures and picks the one that best matches your goals.

1. Your brain's simulations of the future are limited by: A) Nothing — your brain simulates perfectly B) Your past experience — you can't accurately simulate things you've never encountered C) The time of day D) How many friends you have

Answer: B — Your brain remixes past experiences to imagine futures, so gaps in experience lead to less accurate simulations.

1. The **explore vs. exploit dilemma** is: A) A video game challenge B) The trade-off between trying new things and sticking with what you know works C) A test question D) Something only adults deal with

Answer: B — Every plan involves deciding whether to try something new (explore) or stick with what's reliable (exploit).

1. A good rule of thumb for the explore/exploit trade-off is: A) Always explore, never exploit B) Explore more when stakes are low, exploit more when stakes are high C) Never try anything new D) Always stick with what you know

Answer: B — Low-stakes situations are perfect for exploration; high-stakes ones are better for reliable approaches.

1. Why is middle school a great time for exploration? A) Nothing matters in middle school
B) The long-term consequences of most decisions are relatively low, so you can try things and learn
C) You have no other options
D) Exploration is required by law

Answer: B — The relatively low long-term stakes make middle school ideal for trying new things and building experience.

1. The best plans are: A) Rigid and detailed, covering every possible minute
B) Flexible, leaving room for surprise and adaptation
C) Made by someone else
D) Never changed once created

Answer: B — Since you can't predict everything, smart plans leave room to adjust when surprises happen.

1. Instead of planning to write your whole essay on Sunday, a better plan is: A) Don't write the essay at all
B) Break it into parts across multiple days so unexpected events don't ruin everything
C) Write it all Monday morning before class
D) Wait for inspiration to strike

Answer: B — Spreading work across days builds flexibility — if one day goes wrong, you still have progress.

1. The active inference principle for planning is: A) Predict the future perfectly
B) Reduce future uncertainty where you can, and build flexibility where you can't
C) Never plan ahead
D) Only plan for the best case

Answer: B — Good planning is about managing uncertainty, not eliminating it.

1. You can't accurately simulate a school dance you've never attended because: A) Dances are unpredictable by nature
B) Your brain doesn't have enough past experience data to make confident predictions about it
C) Nobody can predict dances
D) Your brain refuses to simulate fun events

Answer: B — Without prior experience, your brain's simulations may be way better or way worse than reality.

1. The "Future Simulation Challenge" asks you to imagine best case, worst case, and most likely case because: A) It's fun to daydream B) Planning based on the most likely case with a backup for the worst case makes goals less stressful C) You should always prepare for the worst D) Best-case scenarios always come true

Answer: B — Running multiple simulations helps you plan realistically while being prepared for problems.

Part B: Short Answer

1. Think of a goal you have for the next month. Describe your "most likely case" simulation and one backup plan for if things don't go perfectly.
2. Give an example of a time you chose to explore (try something new) when the stakes were low. What did you learn from the experience?
3. Describe a situation where a rigid plan fell apart because something unexpected happened. How could a more flexible plan have helped?
4. Using the explore vs. exploit idea, what is one new thing you'd like to try this semester (explore) and one thing you'll stick with because it works (exploit)?
5. Why is planning a weekend hangout with a general structure but flexible details better than planning every minute? How does this connect to managing uncertainty?